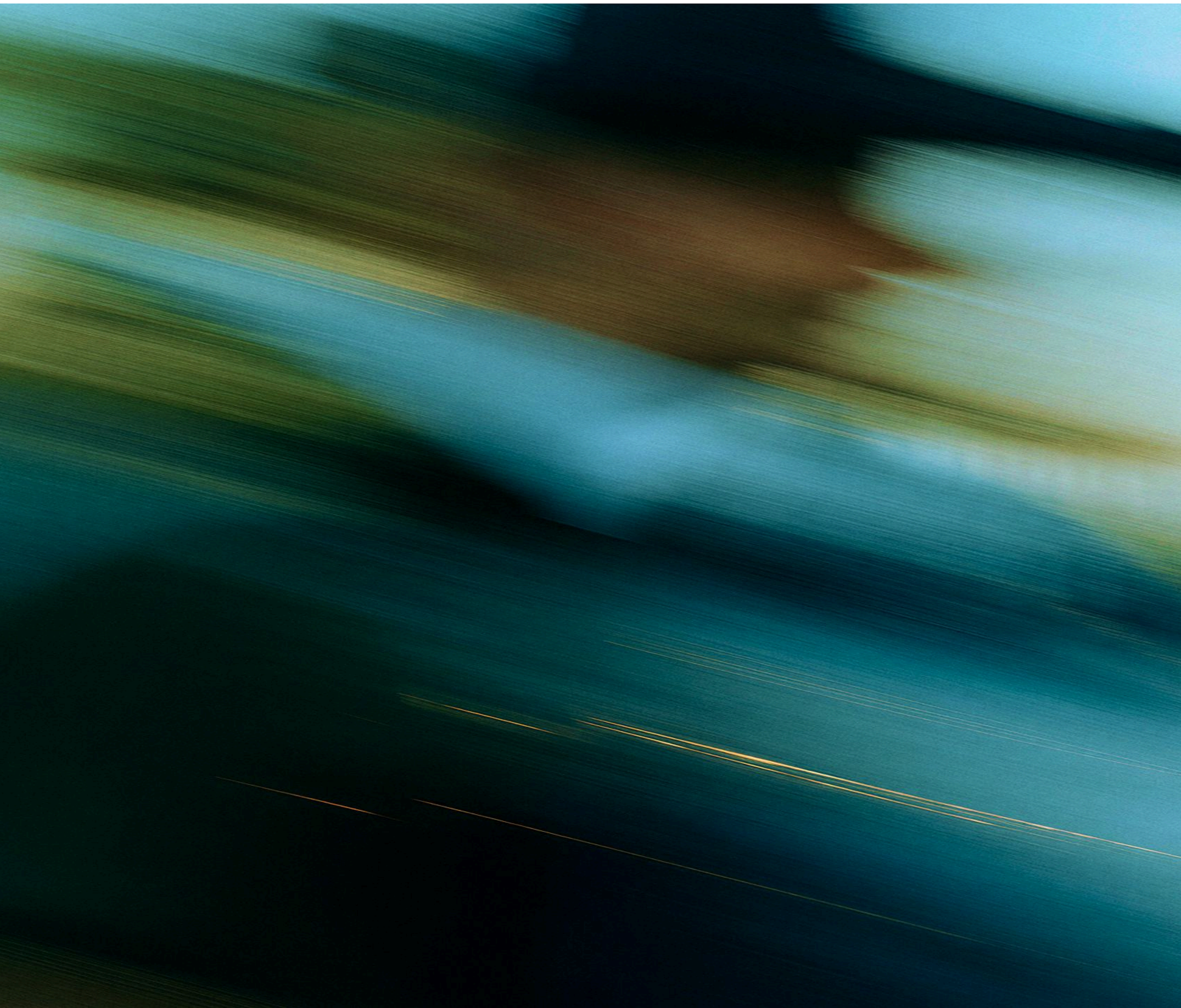
TEMPO

Off-Hours Cash

The Settlement Gap Banks Can Own By Adopting Stablecoins

RESEARCH · BANKING · STABLECOIN SETTLEMENT

TEMPO RESEARCH



Bank-issued stablecoins can unlock off-hours settlement without replacing the deposit franchise.

The opportunity is not retail deposit substitution. It is a tightly bounded wholesale-liquidity product for the hours where the bank deposit rail cannot reach.

Banks can unlock the opportunity to settle cash off-hours by using wholesale deposits as backing for a stablecoin. This can generate new revenue, meet the surge in client demand for 24/7 and weekend settlement, and carry significant potential as a business case.

The picture changes when the liability in question is a wholesale deposit. Unlike core retail deposits, wholesale deposits do not fund the loan book. The LCR assigns them a 40% outflow rate or higher, forcing banks to hold HQLA against them rather than deploy them as credit.

This claim runs counter to the common wisdom that bank-issued stablecoins create a direct cost to the deposit franchise. Under the GENIUS Act, reserves backing a payment stablecoin must be segregated and held in high-quality liquid assets: cash, Treasuries, and central bank reserves.

The opportunity cost of segregating wholesale reserves into stablecoin backing is therefore much lower than it looks when measured against the full deposit franchise. These deposits can be put to work to fund off-hours settlement.

Every dollar a client converts from a deposit into a bank-issued stablecoin is a dollar the bank can no longer lend. The reserve segregation requirement is the structural reason bank-issued stablecoins work against the deposit franchise rather than alongside it.

> A bank-issued stablecoin sleeve allows banks to offer corporate and institutional customers settlement during off-hours. The sleeve is a tightly sized, prefunded, KYC'd pool of stablecoin liquidity that is reversible to deposit form on Monday morning.

Fedwire is closed on Saturdays. CLS is closed on weekends. T2 is also closed on weekends. Roughly $27 trillion sits idle in nostro and vostro accounts globally to pre-fund across those closed windows.

Critically, the weekend has stopped being quiet: in a single April 2026 weekend, trade.xyz processed $1.5 billion in institutional notional, settled in USDC. Fedwire's announced expansion reaches six days a week no earlier than 2028. Saturdays stay closed in practice.

$27T	$1.5B	2028+
ESTIMATED IDLE PRE-FUNDING ACROSS CLOSED WINDOWS	INSTITUTIONAL NOTIONAL OVER ONE APRIL 2026 WEEKEND	EARLIEST FEDWIRE SIX-DAY EXPANSION TIMING

At the right size, the math clears a standalone P&L with meaningful revenue above the cost. The sleeve is not a replacement for the deposit franchise. It is a revenue line that runs alongside it, during hours the franchise cannot reach.

The honest case includes that the sleeve has a definable life, but that window is longer than most banks are modeling. CLS took a decade. The transition to 24/7 central bank money and interoperable tokenized deposits is plausibly ten to fifteen years, constrained by governance rather than technology.

During that window, every weekend, flow routed through a non-bank stablecoin normalizes non-bank money for institutional use. The banks that move in the next eighteen months own the category. The banks that wait discover, five years later, that workflows hardened around someone else.

Dollar settlement still runs inside fixed windows.

There are three areas where those windows fail: nostro account pre-funding, unprotected FX, and off-hours price discovery.

01 Pre-funding

Around $15 to $17 trillion sits in nostro and vostro balances globally, held to cover FX settlement across operating-hour gaps. The capital is trapped by timing: the Asia-Pacific side needs USD finality on Monday morning local time, and the US rail is closed Sunday evening New York time.

02 Unprotected FX

CLS settles an average of $7.2 trillion daily across eighteen currencies and is closed on weekends. In the 2022 BIS Triennial Survey, 31% of global FX turnover, roughly $2.2 trillion daily, settled without PvP protection. Preliminary numbers show unprotected notional reached approximately $4.1 trillion in 2025.

03 Off-hours institutional price discovery

Over a single weekend in April 2026, trade.xyz processed $861.8 million in crude oil notional, $100.7 million in Nasdaq-100, and $202.3 million in S&P 500. Every one of those contracts was margined in USDC.

Market behavior changed first

Geopolitical volatility is accelerating this. When news breaks on a Saturday, positions get adjusted on a Saturday. The venues accommodating that flow are not banks.

The off-hours period is shrinking, but not to zero, and not for a long time. Fedwire is extending to 22 hours a day, six days a week, with implementation no earlier than 2028-2029. Saturdays are not in scope. The CLS weekend closure and the T2 weekend outage both persist. A permanent gap from Friday evening to Sunday evening remains a feature of the dollar settlement day.

Any serious sleeve case starts from why banks instinctively resist stablecoins.

Core deposits are the cheapest, stickiest liability a commercial bank holds. When rates rise, banks have to pay up for wholesale funding almost immediately. Core deposits move slowly, if at all. The gap between the two is where net interest margin lives.

Basel reinforces the preference. The Liquidity Coverage Ratio asks banks to hold enough safe assets to survive a thirty-day stress event, and it assumes different deposit types leak at different speeds.

Stable retail deposits are assumed to leak at low single digits. Corporate deposits tied to operational relationships leak at 25%. Other wholesale funding leaks at 40%, and institutional deposits leak even more.

A bank-issued stablecoin, under the new SCO60 rules that took effect in January 2026, is assumed to leak at 100% unless the bank can continuously identify who holds each unit, which is a demanding operational test.

The legal framework carries the same weight. Deposits are governed by statute. FDIC insurance protects retail balances up to coverage limits. The EBA confirmed in December 2024 that tokenization does not alter the fundamental nature of the deposit claim.

GENIUS reinforces the same distinction in US law by excluding deposits, including tokenized deposits, from the stablecoin definition. The FDIC's April 2026 rulemaking affirmed that tokenized deposits meeting the statutory definition remain deposits under the FDI Act.

Deposits become hotter if they are interchangeable for stablecoins. The franchise is built on sticky money. Once a meaningful share of deposits behaves like a pre-conversion buffer for weekend flow, deposits that looked sticky start acting hot.

BIS Working Paper 1263 documents what happens when a deposit base changes how it treats the bank. During the 2020 market shock, affected Colombian banks saw wholesale deposits drop 27% in days. Even after central bank intervention restored the balances, banks shortened loan maturities and paid more for wholesale funding for quarters afterward.

A stablecoin is not a better bank liability than a deposit. Inside the bank's network, where the deposit claim reaches, the deposit wins. The sleeve is a revenue line that runs alongside the deposit franchise, during hours the franchise cannot reach.

Tokenized deposits preserve the claim, but the network boundary remains.

Deposits settle value between parties that share a bank network. Tokenized deposits extend that capability, for example Kinexsys JP Morgan's Tokenised Deposit Service, while preserving the network boundary. JPMD is a deposit token for institutional clients under OCC interpretive guidance, not a payment stablecoin under GENIUS.

HSBC's product follows the same construction: bank-controlled, permissioned, inside the network. Both work where preserving the deposit claim is the point and the receiving side sits inside the same network. Flows that cross networks still revert to correspondent rails.

Central bank tokenization sits on a different layer. Agora explores how tokenized commercial deposits and wholesale central bank money might share programmable infrastructure. The Eurosystem's Appia and Pontes extend public money into tokenized venues under European rulebooks. These experiments matter over a long horizon. None is a deployable instrument for value crossing perimeters, custodians, venues, and jurisdictions on the timescale of a current product roadmap.

Instrument	Strength	Constraint
Tokenized Deposits	Preserve the deposit claim	Network-Bounded Acceptance
Central Bank Tokenization	Extends the public-money anchor	Governance & Legal-Finality Harmonization
Stablecoins	Transferability across perimeters where accepted	Weaker Underlying Liability than a Deposit

The missing job is moving money across systems that do not talk to each other, in a form that actually settles on arrival. Off-hours settlement is the clearest expression of that missing job today, and the wedge for a broader category.

A stablecoin is a transferable wrapper around safe assets. The wrapper lets the underlying value move across perimeters where the deposit claim cannot. The sleeve occupies the transferability layer.

The design is narrow on purpose.

Friday 4:30 PM NY

Institutional client converts a deposit balance, say $100 million, into a regulated stablecoin under the bank's rulebook.

Weekend Window

Stablecoin moves through institutional rails to post margin, substitute collateral, rebalance exposure, or meet settlement obligations at whitelisted venues.

Monday Morning

Unused balance reverts to the original commercial deposit account via automatic sweep.

The $100 million was a deposit on Friday and a deposit on Monday. For the 48 hours in between, it operated as a regulated stablecoin inside a defined rulebook.

Cap Size	Treasury-approved exposure limits keep incremental LCR and franchise impact immaterial, if any.
Backing	Cash, central bank reserves, short Treasuries, and overnight bank claims. No duration risk over the weekend window.
Fixed Duration	Minted Friday, redeemed Monday. Automatic Monday sweep returns balances to deposit form.
Reporting	Continuous holder identification; real-time position feeds into LCR and NSFR reporting. Without this, SCO60 pushes the product into punitive outflow treatment.
Yield	None. GENIUS Section 4(a)(11) prohibits it, removing the channel through which the sleeve would otherwise cannibalize deposit economics.

Remove any one of these and the instrument stops working as designed. The sleeve is short-duration transactional liquidity inventory, not funding.

A sleeve runs two main cost lines against its fee income.

The largest is LCR opportunity cost. In the worst case, every stablecoin dollar outstanding consumes HQLA at 100% under SCO60. The HQLA that backs it cannot fund loans at a 150-250 bps spread. That lost NIM is the real economic cost the commercial case has to clear.

The second is operating cost: tokenized reserve custody, parallel onchain and off-chain reconciliation, wallet-level sanctions screening, and incident response for chains that never close.

Peak outstanding	$5 billion, Friday evening to Sunday night
Average outstanding	$1.4 billion, time-weighted
LCR opportunity cost	$1.4 billion x 200 bps = $28 million per year
Operating cost	$10-15 million on a lean build
Total annual cost	$38 million
Annualized sleeve flow	$5 billion x 52 weekends = $260 billion
Break-even	1.4-1.6 bps

The sleeve's break-even prices at the top of that band because it bundles three services none of those alternatives carry alone: off-hours execution, HQLA immobilization against a 100% outflow factor, and chain-level operating risk that centralized venues do not assume.

1-3 bps

Tri-party collateral movement charges

5-20 bps

Weekend FX conversion range

5-15 bps

Prime brokerage off- hours financing over base

The sleeve has a definable life, but it is likely measured in years.

The pricing only defends against a client alternative that is materially slower, more expensive on a total-cost basis, or blocked by venue hours or perimeter incompatibility. The first early use case - off-hours institutional collateral, margin, hedging, and portfolio-adjustment flow on weekend-active venues - clears that test cleanly.

Prioritization is Required	The sleeve runs as relationship defense and should not be a core focus product because overuse risks quietly changing behavior in the deposit book it was supposed to protect.
Institution Only	Retail or non-institutional access multiplies compliance cost and breaks the pricing assumption. A bounded sleeve stops being bounded.

Once central bank money extends to 24/7 operation and tokenized deposits reach cross-network interoperability, the structural gap closes. The question is what time horizon that transition has.

CLS, TARGET2, and SEPA each took six to ten years between concept and production. None reconciled multiple central bank legal frameworks at once, which is what cross-perimeter 24/7 settlement in public money requires. The binding constraint is governance, legal-finality harmonization, and inter-central-bank agreements.

CATEGORY RISK

Planning on the fast case while the slow case plays out is how banks cede a category.

During the transition window, every weekend flow routed through a non-bank stablecoin normalizes non-bank money for institutional use. The receiving counterparty builds the workflow around the medium that showed up. Over five years, the default reference for cross-perimeter institutional flows shifts toward non-bank issuance.

Hedge funds, commodity traders, asset managers, and corporate treasuries with exposure to weekend-active venues are already routing around the gap. Each new workflow built on that stack is a workflow that does not default to a bank the next time.

The sleeve requires an inter-bank orchestration layer.

The sleeve design is clear, and the economics beat break-even. The operational load is not trivial: tokenized reserve custody, continuous holder identification, whitelisted destination routing, real-time LCR and NSFR position feeds, automated Monday sweep into deposit form, and chain-level incident response.

Tempo delivers sub-second finality, privacy, native support for compliance policies, and ISO 20022 compatibility for stablecoins, tokenized deposits, and payment systems.

Privacy Zones	Secure, permissioned enclaves on Tempo where client transactions are confidential.
TIP-20 Token Standard & Tempo Policies	Creation of stablecoins or deposit tokens that carry blocklists and allowlists enforced by the network.
Memo Fields	ISO 20022-compatible reference data in onchain transfers for reconciliation and reporting.

Banks, or their consortia, would continue to run the rulebook, holder identification, reserve integration, routing, deposit relationship, and legal frameworks for the sleeve.

Critically, Tempo is designed to be a neutral infrastructure layer, not operated by any single institution. With multiple validators such as Standard Chartered, Visa, and Stripe, the network is already a utility.

The window is open. The economics work at the right size. The banks that move in the next eighteen months own the category.

Selected Source Notes

[1] Keyrock, industry estimates published 2024.

[2] trade.xyz weekend settlement data, April 18-19 2026; Blockworks weekendmarkets.xyz dashboard.

[3] Muller, Ossandon Busch et al., BIS Working Paper 1263, "Deposit runs and bank lending."